

 University of Lancashire	School of Engineering and Computing	Coursework Cover Sheet Interim Report
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Acronyms and Abbreviations

CCMC	Community Coordinated Modelling Centre for Coronal Mass Ejections
CME	Coronal Mass Ejection
EM	Emergency Response (Sector)
FAR	False Alarm Rate
GEO	Geostationary Equatorial Orbit
GIC	Geomagnetic Induced Currents
HSF	Human Space Flight (Sector)
LEO	Low Earth Orbit
MAE	Mean Absolute Error / Mean Arrival-time Error
MEO	Medium Earth Orbit
ML	Machine Learning
NCEI	National Centers for Environmental Information
ND	Neutral Density
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real-Time
NSWP	Numerical Space Weather Prediction
RMSE	Root Mean Squared Error
SEP	Solar Energetic Particle
SF	Solar Flare
STM/C	Space Traffic Management and Coordination (Sector)
SWPC	Space Weather Prediction Center
WSA-ENLIL	Wang-Sheeley-Arge-Enlil CME Supercomputer Model

1 Introduction

1.1 Project Context and Rationale

Spacecraft operations are exposed to hazardous space weather conditions that disrupt satellite electronics. The Space Protection and Intelligent Detection of Environmental Risks (SPIDER) project aims to develop a machine learning system that analyses forecast space weather to assess reliability and provide contextual guidance for satellite operations.

1.2 Report Structure

This report follows the first stages of the project. Section 2 summarises key literature. Section 3 defines the statement of work. Section 4 explains the project planning and management, followed by a risk register in Section 5 and an evaluation of progress to date in Section 6. The Appendices contains useful information to those not familiar with Space Weather.

2 Literature Review

2.1 Overview of Existing Research

Research consistently shows poor accuracy and reliability of space weather forecasts. Operational challenges remain underexplored and uncertainty is not fully reflected in the outputs. (Camporeale & Berger, 2025, p. 1) A verification study found that SWPC's flare forecasts do not outperform baseline models such as persistence and climatology, and exhibit severe calibration issues alongside very high FARs in safety-critical scenarios. (Camporeale & Berger, 2025, p. 1) These weaknesses are evident in the detection of a major storm event after an extended quiet period and when identifying all-clear conditions. Both of which, show a strong bias to over-warning which limits the forecast's value as a source of operational confidence. (Camporeale & Berger, 2025, p. 15)

Broader modelling studies, show similar limitations in CME arrival-time forecasting. Ensemble and physics-based models like the WSA-ENLIL showed a MAE of 10 hours, a systematic early bias of -4 hours and a hit rate of 50%. (Wold, et al., 2018, p. 1) Forecast accuracy varies with changes in input conditions, restricted spacecraft coverage from L1 and GEO coronagraphs, and uncertainty in solar-wind speed and direction. (Riley, et al., 2018, p. 1257) This highlights the need for probabilistic predictions in a non-linear system. (Mays, et al., 2015, p. 1)

Verification studies show that model performance strongly depends on how forecast skill is measured. Owens explains that metrics like MAE and RMSE, are useful baselines but fail when timing errors result in *double penalties*, so a forecast with the correct amplitude but incorrect timing can appear to perform poorly. (Owens, 2018, p. 1859) This limitation is particularly relevant for geomagnetic indices like Kp which respond sharply to changes in the solar wind. (Riley, et al., 2018, p. 1256)

Research studies show a shift towards integrating ML techniques into space weather forecasting, revealing both their potential and constraints. Reviews by

Camporeale and results from DENSER show that supervised learning and neural networks can provide operational enhancements when high quality, well calibrated training data is available. (Camporeale, 2019, p. 1170) (Melcot, et al., 2023, p. 10) Adoption, however, remains slow, partly due to longstanding scepticism within the physics community. (Camporeale, 2019, pp. 1185-1186; 1197) ML has become more accessible, creating opportunities to revisit open challenges and improve existing methods. (Camporeale, 2019, p. 1169) Storm data imbalance and variable NRT data quality further limit performance, and DENSER finds ML prediction models not mature enough to replace physics-based models. (Melcot, et al., 2023, p. 10) Camporeale, suggests the most promising direction is a probabilistic grey-box approach, combining physics-based modelling with ML. (Camporeale, 2019, p. 1)

Alongside the technical aspects, there is also an important operational dimension. The National Survey of User needs revealed that operators request forecast products with greater granularity, explicit uncertainty and reliability indicators to support confident decision-making. Many participants report the current text-based bulletins don't meet the demands of rapid access and automation. As computational approaches evolve, space weather data needs to be more machine-readable and ML-friendly. There's a recognised need to expand application of non-traditional fields in space weather research. (SWAG, 2024, p. 11) STM/C, EM, HSF and Research sectors emphasise the need for a systems-level approach to achieve comprehensive and reliable forecasting. (SWAG, 2024, pp. 12-14)

Across current research, there is broad agreement for continuous evaluation of forecasts. Geospace modelling and ionospheric studies showed performance varies across event types, observational coverage and the solar cycle. (Coster, 2025, p. 138) Uncertainty in boundary conditions, especially solar wind parameters, remains one of the main limitations in NSWP. Regular assessment of prediction skill is therefore essential to ensure operators understand the strengths and limitations of the products they use. (Morley, 2020, p. 21)

2.2 Key Findings and Gaps

Several critical gaps emerged that directly justify the SPIDER concept. First, there is a lack of transparency and reliability in operational verification for SWPC forecasts. Although SWPC provide daily Kp predictions, there is no public product that quantifies it's accuracy or reliability with upstream solar wind conditions. Studies have shown repeatedly that SWPC forecasts have poor calibration and FARs. (Camporeale & Berger, 2025, p. 15) This is a major operational gap for satellite operators, especially in LEO where ND drag modelling depends on these geomagnetic conditions. (SWAG, 2024, p. 78)

Second, forecasts rarely include explicit uncertainty information. While ensemble or physics-based models show a range of possible outcomes, uncertainties are typically not exposed in final broadcast products. This conflicts with survey participant feedback that such information is needed for independent risk assessments. (SWAG, 2024, p. 13)

Third, there is a methodological gap in how forecasts are evaluated. Standard verification metrics are strictly a point-by-point comparison which does not

account for timing uncertainty in NRT data and can penalise models unfairly. (Owens, 2018, p. 1854)

Fourth, space weather prediction has a fundamental challenge due to data imbalance in quiet and storm events and data availability of rare extreme events. (Melcot, et al., 2023, p. 9) ML models tend to overfit to quiet conditions and underperform in extreme events. (Camporeale & Berger, 2025, p. 15)

Fifth, there is a broader disconnect between physics-based forecasting and ML integration, as the literature often trends towards using deep learning to fully replace physics-based models. (Melcot, et al., 2023, p. 5) However, there's room to explore how ML can complement existing forecasting rather than replace them, combining the strengths of both in a grey-box approach. (Camporeale, 2019, p. 1197)

Taken together, the literature supports the need for a tool that provides reliability and uncertainty awareness in geomagnetic forecasting. This will enrich the SPIDER project's development by providing a robust methodological foundation as the project moves towards implementation.

3 Statement of Work (SoW)

3.1 Aim and Objectives

Develop a machine learning model that can improve satellite operator's ability to assess risk and respond intelligently to space weather events.

1. Conduct a thorough literature review, that not only looks at past research but also explores the physical processes of the Sun-Earth system and determines the operational impact.
2. Explore and gather data from public sources, analyse measures and limitations to build a holistic understanding of the data.
3. Prepare and clean the data, engineering the necessary features for dataset creation.
4. Validate datasets with baseline models to assess applicability of Machine Learning.
5. Develop and train a ML model, testing different approaches.
6. Evaluate model performance using a range of analytical techniques to ensure reliability.
7. Develop and evaluate a rule layer to convert model outputs to contextual information for satellite operators.
8. Demonstrate model, evaluate system performance in final report and complete VIVA.

3.2 Background and Justification

Space Weather refers to the effects of the Sun's activity on the space environment around the Earth. The Earth is protected by a magnetic field that shields the planet from most incoming solar energy. During a geomagnetic storm this field is pressed against the Earth allowing that energy to reach deeper layers of the upper atmosphere. This disturbs the ionosphere, a region of atmosphere which contains charged particles and cause the atmospheric density to increase.

(Coster, 2025, p. 135) These changes disrupt communications and tracking across sectors. Satellites can be charged causing unexpected behaviour while astronauts can have higher radiation exposure. (SWAG, 2024, p. 20) These impacts highlight the need for reliable forecasting for operational decision making.

3.3 Planned Activities

Conduct thorough review of literature on the topic of Space Weather and it's operational impact. Perform a data audit, determining what data is available and it's accuracy. Analyse the gathered data into datasets with Python and prepare for modelling. Using baseline modelling, compare performance across metrics. Develop a ML model using scikit-learn and compare with baselines to produce a reliability index (RI). Produce a rule layer that converts RI into contextual recommendations. Evaluate performance with operational requirements. Demonstrate the project.

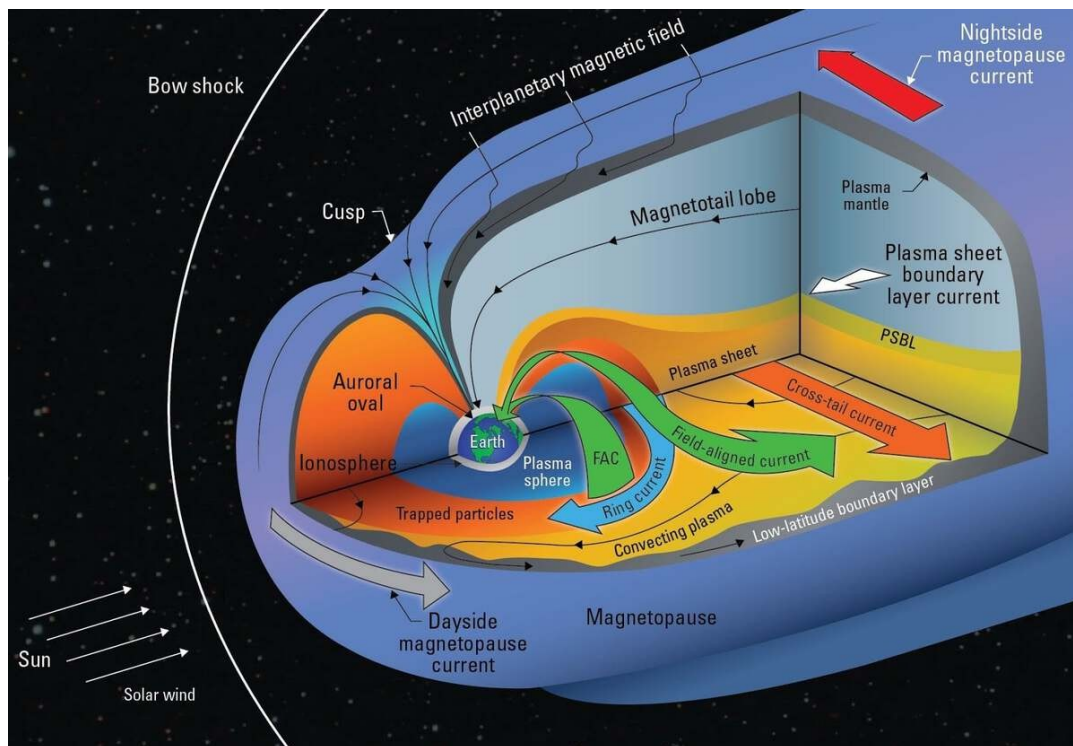


Figure 1: Dynamic environment of the Sun-Earth system. (DrivenData, 2021)

For a more in-depth but clear analysis of the Sun-Earth system and a breakdown of individual phenomena, please see Appendix A.

4 Project Planning and Management

4.1 Work Breakdown Structure (WBS)

R2O2R defines the external strategic context of SPIDER:



Figure 2: The R2O2R Ouroboros

A paired cycle represents Research → Operations → Research. (Morley, 2020, p. 6)

Models that use ML at low-computational cost that conducted thorough verification are encouraged to bridge the gap between research and operations. (Camporeale & Berger, 2025, p. 17)

Within this framework, SPIDER can sit in the research loop during development, testing and evaluation. Once an MVP is produced and thoroughly evaluated against operation needs, it could transition to the operations loop: deployment, testing against new NRT data and feedback from end-users. (Morley, 2020, pp. 6-7)

The internal project structure will follow the CRISPi-DM process (a light-weight student-version of CRISP-DM developed for this project) The need for a simplified variant of CRISP-DM is highlighted by (Schröer, 2021, p. 532) :

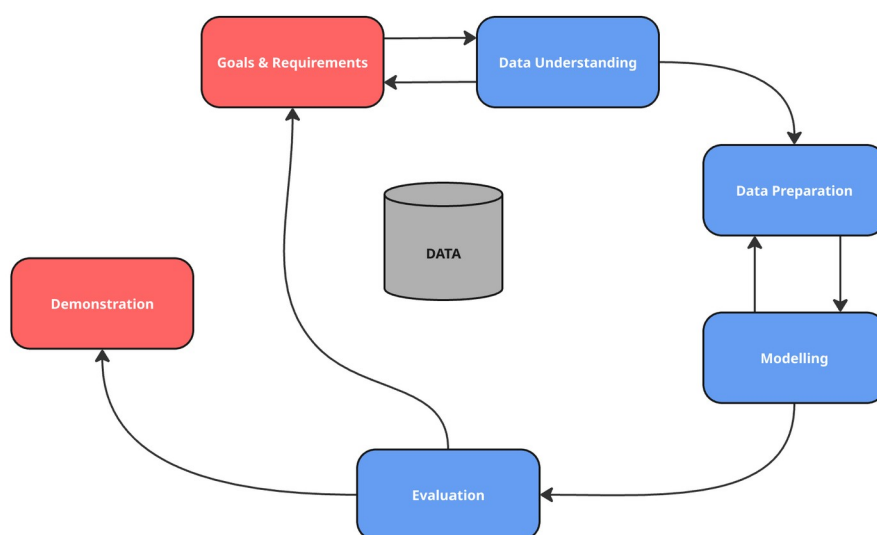


Figure 3: CRISPi-DM streamlines CRISP-DM

In CRISPi-DM, Business Understanding is replaced by Goals and Requirements. Deployment is simplified to Demonstration, and the process follows a semi-iterative approach.

- 1 Goals and Requirements
 - 1.1 Project Start
 - 1.2 Ethics Form
 - 1.3 Literature Review
 - 1.4 Interim Report
 - 1.5 Forecast Verification
 - 1.6 Safing Instructions
 - 1.7 Requirements Analysis
- 2 Data Understanding
 - 2.1 Collect Predicted and Observed Kp
 - 2.2 Collect Upstream Solar Wind (1-hour avg.)
 - 2.3 Additional *Observed* Indices?
 - 2.4 Convert Text to JSON
 - 2.5 *Review against Requirements*
- 3 Data Preparation
 - 3.1 Feature Engineering
 - 3.2 Cleaning and Alignment
 - 3.3 Dataset(s) Creation and Train-Validation Split
 - 3.4 Aggregate JSON to DataFrame
 - 3.5 *Applicability Review After Baseline Modelling*
- 4 Baseline Modelling
 - 4.1 Metrics
 - 4.2 Python Implementation
 - 4.3 Data Visualisation
 - 4.4 *Baseline Review and Return to Data Preparation*
- 5 MVP ML Development
 - 5.1 Select Candidate Models
 - 5.2 Implement Decision Tree Model
 - 5.3 Test Additional Datasets
 - 5.4 Document Python Infrastructure
- 6 Evaluate and Review
 - 6.1 Evaluate MVP Against Baselines
 - 6.2 Test Alternative ML Models
 - 6.3 Check Alignment with Goals & Requirements
 - 6.4 Full System Review and Decision Point
- 7 Demonstration
 - 7.1 Demonstration Model
 - 7.2 Complete Final Report and Poster
 - 7.3 Review against Operational Needs
 - 7.4 Complete VIVA

(Schröer, 2021, p. 531) (Camporeale & Berger, 2025, pp. 16-17)

4.2 Task Specifications and Descriptions

Goals and Literature Review (WBS 1) establish foundations of SPIDER. Review key literature on Sun-Earth system, forecast verification, safing practices and operational requirements. Create requirements analysis to consolidate system goals.

Data Understanding (WBS 2) Collect predicted and observed indices and upstream solar wind to make offline dataset. Metadata definitions and feasibility checks on additional indices. All text datasets will be converted into machine-

readable JSON and reviewed against requirements (WBS 1).

Data Preparation (WBS 3) before modelling by engineering features, cleaning and alignment, forming training and validation splits. Aggregated JSON converted to DataFrame. Review performed after baseline results for ML applicability.

Baseline Modelling (WBS 4) will be implemented using *scikit-learn*. Apply metrics based on (Camporeale & Berger, 2025) methodology. Generate visualisations, assess if feature adjustments required. Return to Data Preparation (WBS 3).

MVP ML Development (WBS 5) after baselines, select model types and implement decision tree. Test generalisation with additional datasets. Ensure modelling infrastructure is documented.

Evaluation and Review (WBS 6) using same metrics as Baselines (WBS 4). Test different approaches and confirm adherence to requirements. Full review decides whether system should iterate (WBS 1) or progress to demonstration.

Demonstration (WBS 7) of working system. Evaluate alignment with operational needs. Prepare final report and undertake VIVA to conclude project (WBS 8).

4.3 Programme (Gantt Chart)

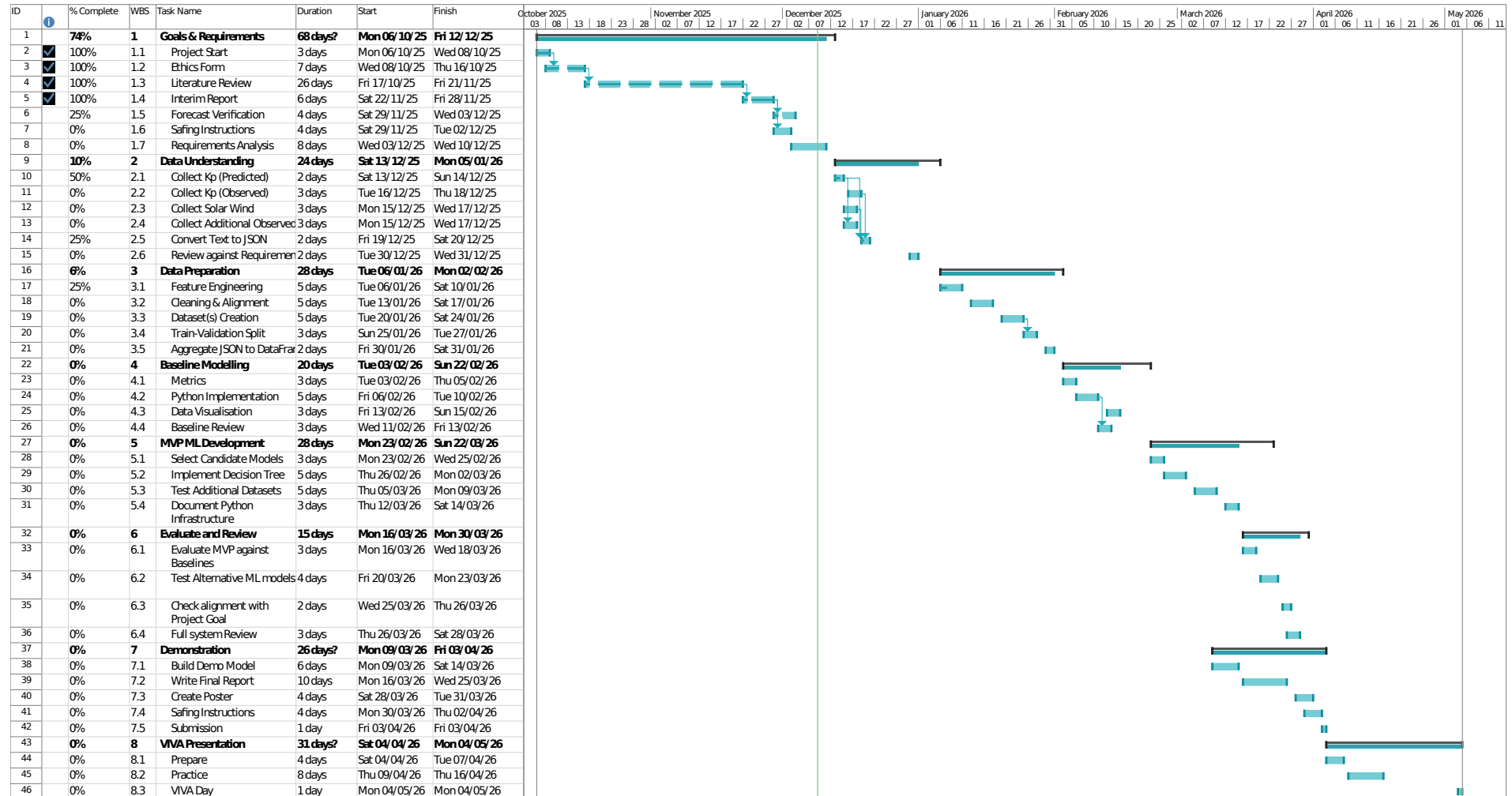


Figure 4: Project progress can be tracked using a Gantt chart

4.5 Dependencies and Interrelationships

Dependencies

- PC access
- Internet access
- Access to SWPC Forecasts, NCEI data archive, NASA OMNI data archive
- GitHub repository access

Interrelationships

- Requirements analysis *needed* to complete first stage.
- Data acquisition *needed* before Data Understanding.
- No Data Preparation *without* Data Understanding, no understanding of the quality or limitations.
- Baseline model *dependent on* Data Preparation.
- MVP model *dependent on* validation of Baseline modelling to assess ML *applicability*.
- Model Evaluation cannot be done *without models to test*.
- No Demonstration *without MVP*.
- Data Understanding informs the rules applied in Data Preparation which in turn shapes structure of final datasets.
- Feature engineering and dataset construction influence Baseline performance which determines *feature refinement*.
- Model evaluation results guide demonstration and reporting. This involves next (potential) cycle of CRISPi-DM methodology.

4.6 Risk Management and Mitigation

1 Identification

Risks are identified throughout the CRISPi-DM methodology. When a new risk occurs it is recorded.

2 Quick Assessment

Evaluate the risk in terms of likelihood and impact. The risk is categorised appropriately to understand it's priority.

3 Mitigation

Based on the priority of the risk, a choice may be made on whether it needs to be acted on immediately or if it is something to be aware of as the project progresses.

4 Risks in Review

Risks are reviewed at each milestone. If a risk increases in severity, then the mitigation is also increased to compensate. This might mean, in extreme cases, a reduction in scope. Certain risks, may require supervisor involvement but ensuring this step is completed on each review, there will be fewer surprises.

5 Integration with GitHub issue tracking

In addition to the formal text based register, risks, bugs, milestones and enhancements will be tracked on the GitHub issue system. The Risk Register will contain a condensed summary of primary risks identified during each review cycle. While GitHub, will maintain the full list of active risks and issues.

4.7 Deliverables and Milestones

#	Deliverables	Due Date
1	Literature Review and Interim Report	12/12/2025
2	Offline Data Repository	05/01/2026
3	MVP Dataset and Feature Set	02/02/2026
4	Baseline Modelling Report	22/02/2026
5	MVP ML Model	22/03/2026
6	Evaluation Summary	30/03/2026
7	Final Demonstration, Report	03/04/2026
8	VIVA Day	04/05/2026

Figure 5: SPIDER deliverables

Milestones	WBS
Complete Goals and Requirements Analysis	1
Data Understanding completed, all datasets stored as JSON.	2
Data Preparation completed and reviewed.	3
Baseline Modelling implemented and reviewed.	4
MVP ML Development completed with additional datasets.	5
Evaluation and Review completed	6
Final Report submitted, Demonstration	7
VIVA completed.	8

Figure 6: SPIDER Milestones

4.8 Documentation and Reporting Structure

Project documentation is maintained through a OneNote notebook for technical ideas, dataset changes and experiments. Critical documents will be written in Word based on CRISPi-DM requirements. Source code will be stored on a private GitHub repository with a commit history. Supervisor will be contacted when milestones are achieved or queries need to be made for specific support. All report materials will be consolidated in final report.

5 Risk Register

Risk	Risk Type	Why it matters?	Likelihood	Impact	Mitigation
Over-Ambition / Scope Creep	Management	Attempting to add more features, models or complexity than the schedule allows.	High	High	Define a strict MVP, with requirements analysis, stick to planned deliverables, escalate ONLY when core elements are complete.
Loss of access to public data	Technical	Could delay dataset creation or affect data continuity.	Medium	High	Keep local copies, contact archive teams to find causes for missing data, conduct data audit.
Data alignment issues	Technical	Lead to incorrect or biased model outputs.	Medium	Medium	Research measurement calculations, follow best practices learnt from literature.
Weak features reduce model performance	ML	Prevent ML from being effective for current task.	Medium	High	Refine feature engineering, iterate with CRISPi-DM.
Late on Deadline	Management	Could result in reduced scope or late deliverables.	Medium	High	Prioritise milestones early, track on Gantt chart.
Data loss or Corruption	Technical	Loss of processed data or scripts.	Low	High	GitHub version control, error checking in code, backups for Key Documents.
Unauthorised access to repo	Security	Could affect integrity of code.	Low	Medium	Restrict repo with SSH authentication, secure development environment.
Health safety concerns from project activities	H&S	Could introduce physical hazards if hardware or lab work was involved.	Very Low	Very Low	Project is fully software based. No hazardous materials or physical equipment involved. Ensure good ergonomics when working only.
Ethical issues relating to sensitive data	Ethical	Could require ethical approval or pose privacy risks.	Very Low	Low	Data sources are public domain historical weather data. No sensitive information is forecast.

Figure 7: Risk Register for standout risks (GitHub Tracker for rest).

6 Progress Evaluation

A detailed review of research across twelve categories was completed to build a clear understanding of the space weather environment. Early inspiration from (SWAG, 2024) began the exploration of NOAA datasets and identified the NCEI as the long-term archive of SWPC forecasts. Examining the SWPC dashboards, it became clear why operators experience challenges when interpreting the displayed information:

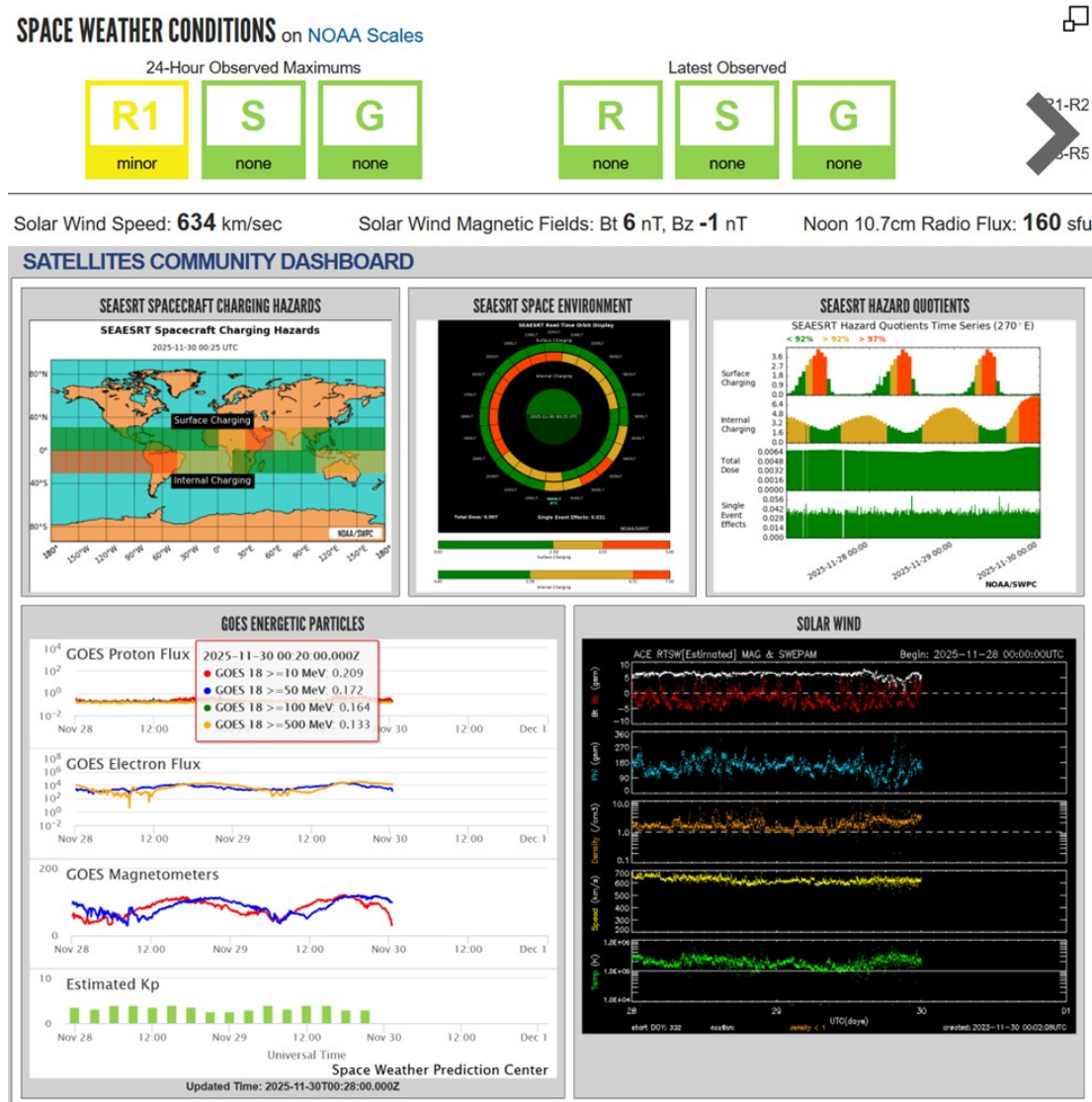


Figure 8: NOAA SWPC Satellite Dashboard (SWPC, NOAA, 2025)

Although, there is scientific detail, it lacks the decision-support that the operators needed. It offers no confidence or reliability information, no indication of operational risk and no guidance. There is no forecast of what will occur here. There is only a snapshot into what *is* occurring.

Initially, solar wind prediction was considered from parameters directly, but it quickly became clear that doing so was computationally expensive.

Deep learning frameworks such as PyTorch and the MATLAB Satellite Communications Toolbox were initially considered. Following the literature review and risk register, their use at this stage was deemed to add unnecessary complexity with little operational benefit. The scope was therefore reduced to focus on meeting objectives through data analysis, reliability assessment and decision-support.

The focus then shifted to Kp. Unlike solar wind, Kp isn't a physical quantity but a logarithmically scaled indicator of global geomagnetic activity. Derived from local K-indices, with the 'p' indicating its planetary scope. Notice on the dashboard, Kp is the only value estimated, assessing the reliability of that prediction became the main focus of the project.

The Kp data was accessible over FTP, so a custom FTP tool was developed to automate downloads by source and data range. While SWPC provides extensive data, forecasts contain gaps. This may be a relatively new product. As of 25/11/2025, the US Government has reopened, an enquiry was submitted to NCEI to clarify the gaps and to request alternative sources where possible.

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Space Protection & Intelligent Detection
of Environmental Risks

Running FTP Access Utility...
Select dataset type (3day / geomag / daypre): 

```

Figure 9: FTP Access Utility for downloading SPIDER datasets

The main forecast and observed products identified for the project are:

1. Geomagnetic Forecast (11/2022-08/2023; 03/2024-Present): Numerical Kp predictions, daily Ap and storm probabilities.
2. Three-Day Forecast (03/2022-08/2023; 03/2024-Present): Numerical Kp values with text-based rationale, solar storm and radio-blackout information.
3. Day Prediction Forecast (12/2013-09/2020;02/2024 - Present) - Local K-indices, useful for validation if gaps can be resolved.
4. Daily Observed Geomagnetic Data (01/1994 - Present) - Observed Kp for comparison.
5. OMNI2 (01/1963 - Present) - Hourly, cross-calibrated solar-wind and IMF data.

Downloaded data must be structured for ML use. JSON was chosen over CSV to maintain flexibility and support metadata. Two parsing scripts have been implemented so far. The *geomag_forecast* parser was straightforward due to its

numeric structure. The *three_day_forecast* parser required additional work to convert rationale into structured metadata. The literature emphasised that inadequate data understanding and preparation are major causes of failure in ML projects. SPIDER will therefore prioritise robust foundational methods.

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Appendices

Appendix A: Space Weather Explained

Space Weather Breakdown from Sol to Earth

Space Weather is the study of the plasma and magnetic dynamics of the Sun-Earth environment. Unlike terrestrial weather, which is created by the Sun's heating of Earth's atmosphere, space weather comes from one dominant source, the Sun. Although the Sun is often viewed as the giver of light and life, it also possesses immense destructive power that humanity has only recognised very recently in cosmological terms. (Morley, 2020, p. 2)

Composed primarily of hydrogen and helium, the Sun, makes up 99.8% of the mass in the solar system and is a massive thermonuclear explosion, held in balance by the inward force of gravity. The scale of the Sun is, astronomical. How fitting! 1.3 million Earth's can fit inside the Sun, if the Earth was a marble, the Sun would be a giant metre-wide beach ball! (McGoodwin, 2009)

The Sun is hot, really hot! It's temperatures reflect its extreme energy: the surface is around 5,505°C, while the core reaches 15,000,000°C! At these temperatures, ordinary matter changes state; metal would melt and granite would vaporise, and within the core, gas becomes plasma where nuclear fusion occurs. Producing an extreme amount of heat, light and energy! (McGoodwin, 2009)

Although the inner workings of the Sun are still an active area of research it is understood that a self-sustaining magnetic dynamo, driven by the motion of hot ionised gas, generates complex magnetic fields that are 1000 times stronger than Earth's magnetic field. Two key components of the magnetic field are the α -effect (poloidal) and the ω -effect (toroidal), that contribute to the formation of sunspots. (Gombosi, 1998) Under certain conditions, the magnetic energy concentrated around these regions can lead to explosive releases of radiation

(solar flares) or, when enough mass is involved coronal mass ejections (CMEs). (Strong, et al., 2017, p. 2388)

Most of this activity is harmless to Earth but occasionally the planet lies directly in the path of this massive ejection of radiation or magnetised plasma. The Sun also produces a steady stream of charged particles known as the solar wind. During a CME, this wind becomes a storm travelling at up to 2200km/s. (Strong, et al., 2017, p. 2394)

As the solar wind reaches Earth, the interplanetary magnetic field (IMF) is forced to divert around the planet's magnetic field, forming a bow shock that slows and heats the flow. The strength of this interaction depends on the IMF vertical direction, with a southward IMF allowing more energy to enter the magnetosphere than a northward one. (Strong, et al., 2017, pp. 2594-2598)

Earth is fortunately shielded by its magnetic field, generated by a smaller internal dynamo in the planet's molten iron core. This magnetosphere acts as a protective barrier, preventing most harmful radiation from reaching the surface and preserving the atmosphere. For much of human history, the visible effects of space weather were known only through the Aurora Borealis and Aurora Australis. While these displays appear beautiful, they also serve as reminders that Earth's magnetic field is protecting the planet from deadly radiation. (Strong, et al., 2017, pp. 2598-2599)

Above Earth's atmosphere, the ionosphere forms a layer of charged particles shaped by the solar radiation. The ionosphere is an important part of radio communication but is sensitive to geomagnetic disturbances. (Coster, 2025, p. 131) During storms, energy builds, forming the ring current, a flow of energetic ions that weaken Earth's magnetic field. (Gombosi, 1998)

As humanity has ventured into space and become increasingly reliant on satellites for communication, navigation and other services, the impacts of space weather have grown more significant. (Morley, 2020, p. 1)

These disturbances can charge and damage satellites, disrupt navigation and communications and increase the neutral density of the upper atmosphere causing drag to satellites in Low-Earth Orbit (LEO). This rise in density, can also alter the trajectories of space debris in unpredictable ways, making accurate tracking more difficult. Astronauts face higher radiation exposure, especially if they are beyond Earth's magnetic protection. On the ground, space weather can interfere with aviation communications, create geomagnetic induced currents (GICs) in power grids and affect sectors using GNSS or precision timing. (SWAG, 2024, pp. 12-14)

Solar Phenomena

Solar Flare (SF): A sudden release of electromagnetic radiation from the Sun, caused by flux in magnetic field lines.

Coronal Mass Ejection (CME): A large eruption of magnetised plasma expelled from the Sun, capable of producing intense geomagnetic storms.

Solar Wind: A steady stream of charged particles flowing out of the Sun's atmosphere, varying in speed and direction based on solar activity.

Interplanetary Magnetic Field (IMF): The magnetic field from the Sun that is carried in a spiral pattern on the solar wind out into space.

Solar Energetic Particles (SEPs): High-energy particles driven by solar activity.

Magnetospheric Parameters

B_t: Total strength (magnitude) of the interplanetary magnetic field.

B_z: The vertical component of the IMF. A southward B_z enhances magnetic reconnection and geomagnetic activity.

Dst index: Measures the strength of the ring current which can indicate the severity of geomagnetic storms.

Ionospheric Parameters

Ring Current: A westward flow of energetic particles around Earth that intensifies during storms and weakens the magnetic field.

Field-Aligned Currents (FACs): Currents that flow between the Magnetosphere and Ionosphere, driving auroral activity.

Auroral Electrojets: Strong ionospheric currents that flow around the auroral oval. They respond rapidly to geomagnetic disturbances.

High Latitude Indices

AE Index: Measures overall electrojet activity and geomagnetic disturbance at higher latitudes.

AU Index: Tracks strength of eastward electrojet.

AL Index: Tracks strength of westward electrojet.

Planetary Indices

K indices: A 3-hour measure of local geomagnetic disturbance, measured at a single ground station.

K_p index: A global, planetary average of K-indices, used to describe overall geomagnetic activity.

A_p index: A linearised version of K_p, that can be averaged.